WORKSHEET

Reducing balance loans

1 Ellie borrows $18 000 to buy a car. She makes monthly repayments of $600 and interest is charged at 7% p.a. which is compounded monthly.

a Write a recurrence relation that gives the amount still owing on the loan after n months.

b How much does Ellie still owe after 3 months?

c How much interest was charged during this period?

2 Liam takes out a business loan of $100 000. Interest is calculated at a rate of 5% p.a. adjusted monthly. He makes repayments of $1500 each month.

a How much interest was charged in the first month after Liam took out the loan?

b Write a recurrence relation that gives the amount Liam still owes on the loan after n months.

c How much does Liam owe after 1 year?

3 Lucy borrows $2000. She repays $200 each month and is charged 12% interest p.a. which is compounded monthly.

a How much interest will Lucy pay in the first 2 months?

b Express the amount Lucy owes on the loan after n months as a recurrence relation.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00086.indd 1 16/09/15 11:40 AM

c How long will it take Lucy to repay the loan?

4 Ned borrows money to buy a motorbike. The bike costs $29 000, but Ned borrows an extra $1000 to cover insurance and registration costs. Interest is charged at a rate of 7.4% p.a. compounded monthly. Ned makes monthly repayments of $861.

a How much interest is Ned charged in the second month after taking out the loan?

b Write a recurrence relation to express the amount Ned owes after n months.

5 Deran borrows $5000 from his friend Crispin. Crispin charges Deran 0.5% interest per month on the amount owing, and Deran makes monthly repayments of $175.

a Write a recurrence relation to express the amount Deran owes Crispin after n months.

b How much will Deran owe after 6 months?

6 Bradley borrows $12 000 from a credit union. Interest is compounded quarterly and repayments are made quarterly. The interest rate is 8.4% p.a. and Bradley's repayments are $800 per quarter.

a How much will Bradley owe after the first quarter?

b Write a recurrence relation to express the amount Bradley owes after n quarters.

c How long will it take Bradley to repay the loan?

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00086.indd 2 16/09/15 11:40 AM

7 The recurrence relation $A_n = \left(1 + \frac{9}{1200}\right)A_{n-1} - 525$ where $A_0 = \$35\,000$ expresses the amount owing on a loan after n months.

a How much money was borrowed?

b What is the interest rate charged per annum?

8 Gandalf takes out a loan for a new house. The recurrence relation:

$A_n = \left(1 + \frac{4.25}{1200}\right)A_{n-1} - 3268$ where $A_0 = \$210\,000$ expresses the amount Gandalf owes on the loan after n months.

a How much money did Gandalf borrow?

b What was the interest rate per annum?

9 Ghezel borrows money to pay for a holiday. The amount she owes on the loan after n months is expressed by the recurrence relation:

$A_n = \left(1 + \frac{9.75}{1200}\right)A_{n-1} - 850$ where $A_0 = \$25\,000$.

a How much are Ghezel's monthly repayments?

b What is the interest rate charged per month?

10 Luisa borrows \$16 000. She pays 0.75% compound interest per month, and repays \$450 each month. Plot a graph showing the amount owing on Luisa's loan over the first year.

© Cengage Learning Australia Pty Ltd 2015

MATHS12WK00086.indd 3 16/09/15 11:40 AM

Answers

1 **a** $A_n = \left(1 + \frac{7}{1200}\right)A_{n-1} - 600$ where $A_0 = \$18\,000$

b \$16 506.32

c \$306.32

2 **a** \$416.67

b $A_n = \left(1 + \frac{5}{1200}\right)A_{n-1} - 1500$ where $A_0 = \$100\,000$

c \$86 697.91

3 **a** \$38.20

b $A_n = \left(1 + \frac{12}{1200}\right)A_{n-1} - 200$ where $A_0 = \$2000$

c 11 months

4 **a** \$180.83

b $A_n = \left(1 + \frac{7.4}{1200}\right)A_{n-1} - 861$ where $A_0 = \$30\,000$

5 **a** $A_n = \left(1 + \frac{0.5}{100}\right)A_{n-1} - 175$ where $A_0 = \$5000$

b \$4088.68

6 **a** \$11 452

b $A_n = \left(1 + \frac{8.4}{400}\right)A_{n-1} - 800$ where $A_0 = \$12\,000$

c 4 years and 9 months

7 **a** \$35 000

b 9%

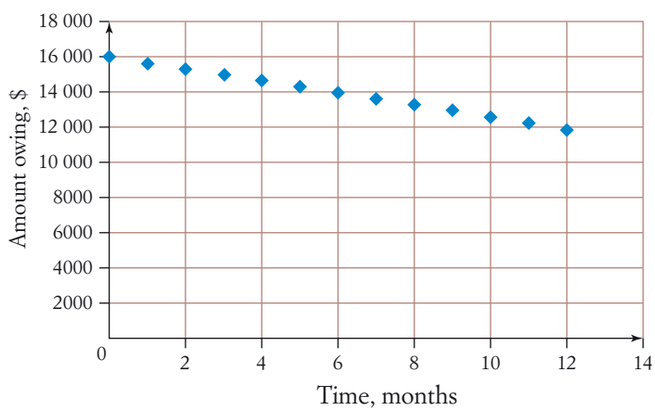
8 **a** \$210 000

b 4.25%

9 **a** \$850

b 0.8125%

10



© Cengage Learning Australia Pty Ltd 2015